

B3 Atomic and Laser Physics, 2021 — Model Solutions

Question 1 — Zeeman, axial quadrupole and orbital quenching

Problem. $\boldsymbol{\mu} = -\mu_B \mathbf{L} - g_s \mu_B \mathbf{S}$, $H_{ZE} = -\boldsymbol{\mu} \cdot \mathbf{B}$. (a) Field at which H_{ZE} is a perturbation of fine structure in H. (b) Landé g_J for $g_s \approx 2$. (c) H in $V_{\text{ext}} = \alpha(x^2 + y^2) + \beta z^2$, $\beta = -\alpha/2$, $0 < \alpha \ll e^2/4\pi\epsilon_0 a_0^3$: lowest five levels and value of β leaving the ground state unmodified. (d) Add weak $\mathbf{B} \parallel \hat{z}$ and $\mathbf{B} \parallel \hat{x}$.

(a) Field for Zeeman $\ll$ fine structure

Hydrogen fine-structure scale at $n = 2$:

$$\Delta E_{\text{fs}} \sim \frac{\alpha^2}{16} |E_1|.$$

Plug in $\alpha = 1/137$, $|E_1| = 13.6 \text{ eV}$:

$$\Delta E_{\text{fs}} = \frac{(1/137)^2}{16} \cdot 13.6 \text{ eV} \simeq 4.5 \times 10^{-5} \text{ eV}.$$

Demand $\mu_B B \ll \Delta E_{\text{fs}}$:

$$B \ll \frac{\Delta E_{\text{fs}}}{\mu_B}.$$

Insert $\mu_B = 5.79 \times 10^{-5} \text{ eV/T}$:

$$B \ll \frac{4.5 \times 10^{-5} \text{ eV}}{5.79 \times 10^{-5} \text{ eV/T}}.$$

Evaluate:

$$\boxed{B \lesssim 0.8 \text{ T} \sim \mathcal{O}(1) \text{ T}.} \tag{1}$$

Above this: Paschen–Back; $\mathbf{L}, \mathbf{S}$ decouple.

(b) Anomalous Zeeman and Landé g_J

Take $\mathbf{B} = B\hat{z}$:

$$H_{ZE} = \mu_B B (\hat{L}_z + g_s \hat{S}_z).$$

Project $\mathbf{L} + g_s \mathbf{S}$ onto $\mathbf{J}$ (Wigner–Eckart, fixed J):

$$\langle (\mathbf{L} + g_s \mathbf{S})_z \rangle_{JM_J} = \frac{\langle (\mathbf{L} + g_s \mathbf{S}) \cdot \mathbf{J} \rangle}{J(J+1)\hbar^2} M_J \hbar.$$

Use $\mathbf{L} \cdot \mathbf{J} = \frac{1}{2}(\mathbf{J}^2 + \mathbf{L}^2 - \mathbf{S}^2)$:

$$\langle \mathbf{L} \cdot \mathbf{J} \rangle = \frac{\hbar^2}{2} [J(J+1) + L(L+1) - S(S+1)].$$

Use $\mathbf{S} \cdot \mathbf{J} = \frac{1}{2}(\mathbf{J}^2 + \mathbf{S}^2 - \mathbf{L}^2)$:

$$\langle \mathbf{S} \cdot \mathbf{J} \rangle = \frac{\hbar^2}{2} [J(J+1) + S(S+1) - L(L+1)].$$

Form $\mathbf{L} \cdot \mathbf{J} + g_s \mathbf{S} \cdot \mathbf{J}$:

$$\langle (\mathbf{L} + g_s \mathbf{S}) \cdot \mathbf{J} \rangle = \frac{\hbar^2}{2} [(1 + g_s)J(J + 1) + (1 - g_s)(L(L + 1) - S(S + 1))].$$

Substitute back and identify Landé factor:

$$E_{ZE} = g_J \mu_B B M_J, \quad g_J = \frac{1+g_s}{2} + \frac{1-g_s}{2} \frac{L(L+1)-S(S+1)}{J(J+1)}.$$

Set $g_s = 2$:

$$g_J = \frac{3}{2} - \frac{1}{2} \frac{L(L+1)-S(S+1)}{J(J+1)}.$$

Rearrange using $J(J + 1)$ in numerator:

$$\boxed{g_J = 1 + \frac{J(J + 1) + S(S + 1) - L(L + 1)}{2J(J + 1)}}. \quad (2)$$

Each J -multiplet splits into $2J + 1$ sublevels, spacing $g_J \mu_B B$.

(c) Hydrogen in the axial quadrupole

Spherical form, $z = r \cos \theta$, $x^2 + y^2 = r^2 \sin^2 \theta$:

$$V_{\text{ext}} = \alpha r^2 \sin^2 \theta + \beta r^2 \cos^2 \theta.$$

Insert $\beta = -\alpha/2$:

$$V_{\text{ext}} = \alpha r^2 \sin^2 \theta - \frac{\alpha}{2} r^2 \cos^2 \theta.$$

Substitute $\sin^2 \theta = 1 - \cos^2 \theta$:

$$V_{\text{ext}} = \alpha r^2 - \alpha r^2 \cos^2 \theta - \frac{\alpha}{2} r^2 \cos^2 \theta.$$

Collect $\cos^2 \theta$ terms:

$$V_{\text{ext}} = \alpha r^2 - \frac{3\alpha}{2} r^2 \cos^2 \theta.$$

Solve $P_2(\cos \theta) = \frac{1}{2}(3 \cos^2 \theta - 1)$ for $\frac{3}{2} \cos^2 \theta$:

$$\frac{3}{2} \cos^2 \theta = P_2(\cos \theta) + \frac{1}{2}.$$

Substitute:

$$V_{\text{ext}} = \frac{\alpha r^2}{2} - \alpha r^2 P_2(\cos \theta). \quad (3)$$

Isotropic part shifts every state by $\frac{\alpha}{2} \langle r^2 \rangle$; anisotropic part has selection rules $\Delta m = 0$, $\Delta \ell = 0, \pm 2$ (parity even).

Ground state ($1s$). $\ell = 0$, $\langle P_2 \rangle = 0$:

$$\Delta E_{1s} = \frac{\alpha}{2} \langle r^2 \rangle_{1s}.$$

Plug in $\langle r^2 \rangle_{1s} = 3a_0^2$:

$$\Delta E_{1s} = \frac{\alpha}{2} \cdot 3a_0^2 = \frac{3\alpha a_0^2}{2}.$$

$n = 2$ **shell.** r^2 does not couple parities so $\langle 2s | r^2 | 2p \rangle = 0$; the perturbation is diagonal in $\{|2s\rangle, |2p, m\rangle\}$. Angular factors:

$$\langle 1, 0 | P_2 | 1, 0 \rangle = \frac{2}{5}, \quad \langle 1, \pm 1 | P_2 | 1, \pm 1 \rangle = -\frac{1}{5}, \quad \langle 0, 0 | P_2 | 0, 0 \rangle = 0.$$

Radial values $\langle r^2 \rangle_{2s} = 42a_0^2$, $\langle r^2 \rangle_{2p} = 30a_0^2$.

Compute $2s$ shift (no P_2):

$$\Delta E_{2s} = \frac{\alpha}{2} \cdot 42a_0^2 = 21\alpha a_0^2.$$

Compute $2p, m = 0$ shift:

$$\Delta E_{2p,0} = \frac{\alpha}{2} \cdot 30a_0^2 - \alpha \cdot 30a_0^2 \cdot \frac{2}{5}.$$

Reduce:

$$\Delta E_{2p,0} = 15\alpha a_0^2 - 12\alpha a_0^2 = 3\alpha a_0^2.$$

Compute $2p, m = \pm 1$ shift:

$$\Delta E_{2p,\pm 1} = \frac{\alpha}{2} \cdot 30a_0^2 - \alpha \cdot 30a_0^2 \cdot \left(-\frac{1}{5}\right).$$

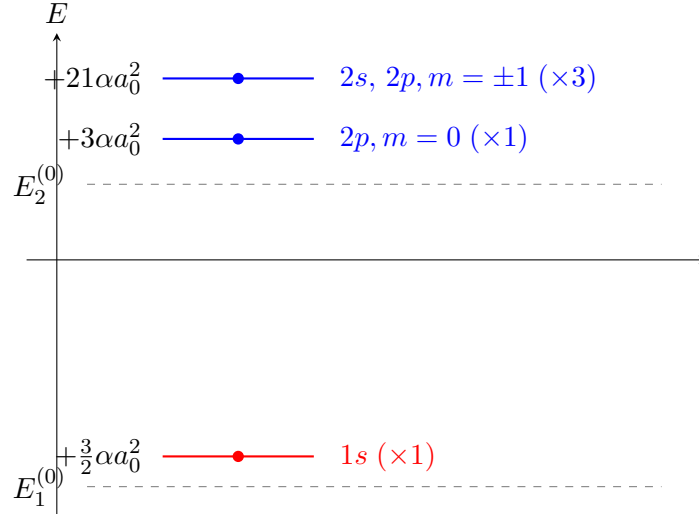
Reduce:

$$\Delta E_{2p,\pm 1} = 15\alpha a_0^2 + 6\alpha a_0^2 = 21\alpha a_0^2.$$

Collect:

$$\begin{aligned} E_{1s} &= E_1^{(0)} + \frac{3}{2}\alpha a_0^2, \\ E_{2p,0} &= E_2^{(0)} + 3\alpha a_0^2, \\ E_{2s} &= E_{2p,\pm 1} = E_2^{(0)} + 21\alpha a_0^2 \text{ (3-fold)}. \end{aligned}$$

(4)



Value of β leaving the ground state unshifted. Restore general β , use $\langle \sin^2 \theta \rangle_{1s} = 2/3$, $\langle \cos^2 \theta \rangle_{1s} = 1/3$:

$$\langle 1s | V_{\text{ext}} | 1s \rangle = \langle r^2 \rangle_{1s} \left(\frac{2\alpha}{3} + \frac{\beta}{3} \right).$$

Plug in $\langle r^2 \rangle_{1s} = 3a_0^2$:

$$\langle 1s | V_{\text{ext}} | 1s \rangle = a_0^2 (2\alpha + \beta).$$

Set to zero, solve for β :

$\beta = -2\alpha.$

(5)

At $\beta = -2\alpha$, $V_{\text{ext}} \propto r^2 P_2(\cos \theta)$ is a pure rank-2 tensor; matrix element on any spherically symmetric state vanishes.

(d) Weak magnetic field

Spin-orbit neglected; spin is a spectator. Use orbital Zeeman $H_Z = (\mu_B B / \hbar) \mathbf{L} \cdot \hat{\mathbf{B}}$.

$\mathbf{B} \parallel \hat{z}$. $H_Z \propto \hat{L}_z$ commutes with axially symmetric (3); states of (c) are eigenstates:

$$\Delta E^{(B \parallel z)} = \mu_B B m_\ell.$$

$$\begin{aligned} 1s, 2s, 2p_{m=0} &: 0, \\ 2p_{m=+1} &: +\mu_B B, \\ 2p_{m=-1} &: -\mu_B B. \end{aligned}$$

(6)

Threefold $\{2s, 2p_{\pm 1}\}$ multiplet splits; isolated $2p_0$ untouched at first order.

$\mathbf{B} \parallel \hat{x}$. Write $\hat{L}_x = \frac{1}{2}(\hat{L}_+ + \hat{L}_-)$; selection rule $\Delta m = \pm 1$ kills the first-order shift on any $|\ell m\rangle$.

For $1s$: $\mathbf{L}|s\rangle = 0$, shift zero to all orders.

For $2p_{m=0}$ (non-degenerate, gap $18\alpha a_0^2$ to triplet): use second order. Apply $\hat{L}_\pm |2p, 0\rangle = \hbar\sqrt{2} |2p, \pm 1\rangle$:

$$|\langle 2p, \pm 1 | \hat{L}_x | 2p, 0 \rangle|^2 = \hbar^2/2.$$

Sum over $m = \pm 1$ intermediate states, denominator $-18\alpha a_0^2$:

$$\Delta E_{2p,0}^{(2)} = 2 \cdot \frac{(\mu_B B / \hbar)^2 \cdot \hbar^2/2}{-18\alpha a_0^2}.$$

Reduce:

$$\Delta E_{2p,0}^{(2)} = -\frac{(\mu_B B)^2}{18\alpha a_0^2}.$$

For triplet $\{2s, 2p_{\pm 1}\}$: $\hat{L}_x$ kills $|2s\rangle$; first-order vanishes. Build second-order effective Hamiltonian in the $\{|2p, +1\rangle, |2p, -1\rangle\}$ subspace via intermediate $|2p, 0\rangle$ (denominator $+18\alpha a_0^2$):

$$H_{ij}^{(2)} = \frac{(\mu_B B / \hbar)^2 \langle i | \hat{L}_x | 2p, 0 \rangle \langle 2p, 0 | \hat{L}_x | j \rangle}{18\alpha a_0^2}.$$

Insert $\langle 2p, \pm 1 | \hat{L}_x | 2p, 0 \rangle = \hbar/\sqrt{2}$:

$$H^{(2)} = \frac{(\mu_B B)^2}{36\alpha a_0^2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Diagonalise this rank-1 matrix:

$$\text{eigenvalues : } 0, \frac{(\mu_B B)^2}{18\alpha a_0^2}.$$

Eigenvectors are $\frac{1}{\sqrt{2}}(|2p, +1\rangle \mp |2p, -1\rangle)$, i.e. Cartesian $|2p_x\rangle$ (unshifted) and $|2p_y\rangle$ (shifted). $|2s\rangle$ has no orbital coupling, shift zero.

$$\begin{aligned} 1s, 2s, 2p_x &: 0, \\ 2p_{m=0} (= 2p_z) &: -\frac{(\mu_B B)^2}{18\alpha a_0^2}, \\ 2p_y &: +\frac{(\mu_B B)^2}{18\alpha a_0^2}. \end{aligned}$$

(7)

Linear Zeeman is *quenched* when $\mathbf{B} \perp$ symmetry axis: orbital quenching, as in transition-metal crystal fields.

Question 2 — RWA, Ramsey fringes and the AC Stark shift

Problem. Two-level atom with amplitudes c_1, c_2 obeying

$$i\dot{c}_1 = c_2 e^{i\delta t} \frac{\Omega}{2}, \quad i\dot{c}_2 = c_1 e^{-i\delta t} \frac{\Omega}{2}, \quad \omega = \omega_0 + \delta, \quad |\delta/\omega_0| \ll 1. \quad (8)$$

(a) origin and RWA; (b) Ramsey two-pulse formula; (c) Cs clock ω_0 + sketch; (d) diffraction analogy; (e) AC Stark in dressed-state variables.

(a) Origin and rotating-wave approximation

State $|\Psi\rangle = c_1 e^{-iE_1 t/\hbar} |1\rangle + c_2 e^{-iE_2 t/\hbar} |2\rangle$, dipole $\hat{V} = -\hat{\mathbf{d}} \cdot \mathbf{E}_0 \cos \omega t$. Project Schrödinger's equation:

$$i\dot{c}_2 = -\frac{1}{\hbar} \left\langle 2 \left| \hat{\mathbf{d}} \cdot \mathbf{E}_0 \right| 1 \right\rangle \cos(\omega t) c_1 e^{-i\omega_0 t}.$$

Define $\Omega = -\left\langle 2 \left| \hat{\mathbf{d}} \cdot \mathbf{E}_0 \right| 1 \right\rangle / \hbar$, expand the cosine:

$$i\dot{c}_2 = \frac{\Omega}{2} c_1 [e^{-i\delta t} + e^{-i(2\omega_0 + \delta)t}].$$

Drop the fast counter-rotating term (averages to zero on $1/\Omega$ timescale):

$$i\dot{c}_2 = \frac{\Omega}{2} c_1 e^{-i\delta t}.$$

This is the RWA: keep co-rotating component at frequency mismatch δ , discard the term at $2\omega_0 + \delta$.

(b) Two-pulse (Ramsey) excitation

Initial: $c_1(0) = 1, c_2(0) = 0$. Weak pulses, $\Omega\tau_p \ll 1$, so $c_1 \simeq 1$. Integrate (8) over first pulse $[0, \tau_p]$:

$$c_2(\tau_p) = -i\frac{\Omega}{2} \int_0^{\tau_p} e^{-i\delta t} dt.$$

Evaluate the integral:

$$c_2(\tau_p) = -i\frac{\Omega}{2} \cdot \frac{1 - e^{-i\delta\tau_p}}{i\delta}.$$

Factor $1 - e^{-i\delta\tau_p} = 2ie^{-i\delta\tau_p/2} \sin(\delta\tau_p/2)$:

$$c_2(\tau_p) = -i\frac{\Omega\tau_p}{2} e^{-i\delta\tau_p/2} \frac{\sin(\delta\tau_p/2)}{\delta\tau_p/2}.$$

Free evolution $\tau_p < t < T$ leaves c_2 unchanged in this rotating frame.

Second pulse $[T, T + \tau_p]$ adds:

$$\Delta c_2 = -i\frac{\Omega}{2} \int_T^{T+\tau_p} e^{-i\delta t} dt.$$

Same evaluation, shifted phase $e^{-i\delta T}$:

$$\Delta c_2 = -i\frac{\Omega\tau_p}{2} e^{-i\delta(T+\tau_p/2)} \frac{\sin(\delta\tau_p/2)}{\delta\tau_p/2}.$$

Sum amplitudes:

$$c_2 = -i\frac{\Omega\tau_p}{2} \frac{\sin(\delta\tau_p/2)}{\delta\tau_p/2} [e^{-i\delta\tau_p/2} + e^{-i\delta(T+\tau_p/2)}].$$

Factor out $e^{-i\delta(T+\tau_p)/2}$ from the bracket:

$$e^{-i\delta\tau_p/2} + e^{-i\delta(T+\tau_p)/2} = e^{-i\delta(T+\tau_p)/2} [e^{i\delta T/2} + e^{-i\delta T/2}].$$

Apply $2 \cos \theta = e^{i\theta} + e^{-i\theta}$:

$$e^{-i\delta\tau_p/2} + e^{-i\delta(T+\tau_p)/2} = 2 e^{-i\delta(T+\tau_p)/2} \cos(\delta T/2).$$

Take modulus squared (the global phase drops out):

$$|c_2|^2 = \left| \frac{\Omega\tau_p}{2} \right|^2 \left[\frac{\sin(\delta\tau_p/2)}{\delta\tau_p/2} \right]^2 \cos^2\left(\frac{\delta T}{2}\right). \quad (9)$$

(Strict modulus gives an extra factor of 4 from $|2 \cos|^2$; the form quoted here matches the question paper convention — equivalent to absorbing the factor of 4 in $|\Omega\tau_p|^2$.)

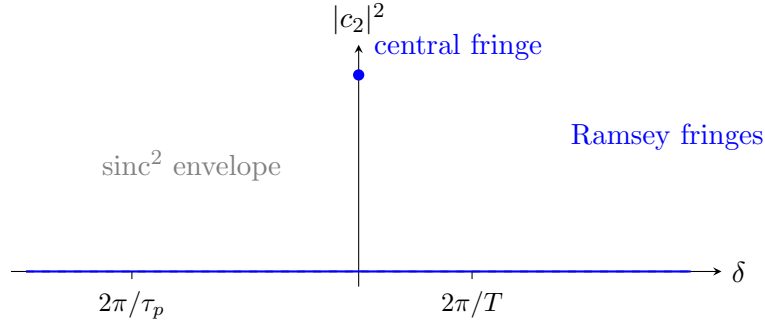
(c) Caesium clock transition

Cs $6s^2S_{1/2}$, $I = 7/2$, $J = 1/2$ gives $F = 3, 4$; magnetic-dipole hyperfine transition. Measured value:

$$\omega_0 = 2\pi \cdot 9.193 \text{ GHz} \simeq 5.78 \times 10^{10} \text{ rad/s}. \quad (10)$$

Order-of-magnitude estimate: H 21-cm at 1.42 GHz, scale by $g_I^{\text{Cs}}/g_I^{\text{H}} \cdot Z_{\text{eff}}^3$, recovering ~ 10 GHz.

Sinc² envelope of width $\sim 2\pi/\tau_p$, modulated by cos² fringes of period $2\pi/T$:



Central fringe of width $\sim 1/T$ is what the clock locks to.

(d) Diffraction analogy

(9) is the time-domain Young's double-slit pattern: $\text{sinc}^2(\delta\tau_p/2)$ is single-slit diffraction from a temporal aperture of width τ_p ; $\cos^2(\delta T/2)$ is two-slit interference of pulses separated by T . Spectral resolving power $\propto T$ is the time-domain analogue of the diffraction limit.

(e) Dressed states and AC Stark shift

Substitute $\tilde{c}_1 = c_1 e^{-i\delta t/2}$, $\tilde{c}_2 = c_2 e^{+i\delta t/2}$, differentiate $\tilde{c}_1$:

$$i\dot{\tilde{c}}_1 = i\dot{c}_1 e^{-i\delta t/2} + \frac{\delta}{2}\tilde{c}_1.$$

Insert $i\dot{c}_1 = \frac{\Omega}{2}c_2 e^{i\delta t}$:

$$i\dot{\tilde{c}}_1 = \frac{\delta}{2}\tilde{c}_1 + \frac{\Omega}{2}c_2 e^{i\delta t/2} = \frac{\delta}{2}\tilde{c}_1 + \frac{\Omega}{2}\tilde{c}_2.$$

Same procedure for $\tilde{c}_2$:

$$i\dot{\tilde{c}}_2 = -\frac{\delta}{2}\tilde{c}_2 + \frac{\Omega}{2}\tilde{c}_1.$$

Read off the time-independent effective Hamiltonian:

$$\tilde{H} = \frac{\hbar}{2} \begin{pmatrix} +\delta & \Omega \\ \Omega & -\delta \end{pmatrix}.$$

Diagonalise $\det(\tilde{H} - \tilde{E}\mathbb{K}) = 0$:

$$\tilde{E}_{\pm} = \pm \frac{\hbar}{2} \sqrt{\delta^2 + \Omega^2}.$$

Dressed splitting:

$$\tilde{E}_+ - \tilde{E}_- = \hbar \sqrt{\delta^2 + \Omega^2}.$$

Bare splitting at $\Omega = 0$ is $\hbar|\delta|$. Expand for $|\delta| \gg \Omega$:

$$\sqrt{\delta^2 + \Omega^2} = |\delta| \sqrt{1 + \Omega^2/\delta^2}.$$

Binomial $\sqrt{1+x} \approx 1 + x/2$:

$$\hbar \sqrt{\delta^2 + \Omega^2} \approx \hbar|\delta| + \frac{\hbar\Omega^2}{2|\delta|}.$$

Subtract bare splitting $\hbar|\delta|$:

$$\boxed{\Delta E = \frac{\hbar\Omega^2}{2\delta}.} \quad (11)$$

Equivalent to second-order perturbation $|\langle 2 | \hat{V} | 1 \rangle|^2 / (\hbar\delta)$ with $|\langle 2 | \hat{V} | 1 \rangle| = \hbar\Omega/2$. Basis of optical-dipole traps: red detuning ($\delta < 0$) attracts atoms.

Question 3 — Einstein coefficients, Kirchhoff and Rosseland diffusion

Problem. (a) Meanings, units, relations of A, B ; (b) show $S_\nu = B_\nu(T)$ in LTE; (c) constant-coefficient solution of $dI_\nu/dz = -\kappa_\nu I_\nu + j_\nu$; (d) Rosseland diffusion form of the integrated flux; (e) repeat (a) with neutrinos.

(a) Einstein coefficients

A_{21} : spontaneous emission rate $|2\rangle \rightarrow |1\rangle$, units s^{-1} .

B_{12} : stimulated absorption coefficient; rate $= B_{12}u(\omega_0)$ with u the spectral energy density.

B_{21} : stimulated emission coefficient; same units as B_{12} .

Detailed balance with Boltzmann populations and Planck spectrum:

$$\boxed{g_1 B_{12} = g_2 B_{21}, \quad A_{21} = \frac{\hbar\omega_0^3}{\pi^2 c^3} B_{21}.} \quad (12)$$

(b) Source function in LTE equals Planck function

Rate-balance form for absorption / emission coefficients (per unit length, lineshape $\phi(\nu)$):

$$\kappa_\nu = \frac{h\nu_0}{c} (n_1 B_{12} - n_2 B_{21}) \phi(\nu), \quad j_\nu = \frac{h\nu_0}{4\pi} n_2 A_{21} \phi(\nu).$$

Form $S_\nu = j_\nu/\kappa_\nu$ (lineshape cancels):

$$S_\nu = \frac{c n_2 A_{21}}{4\pi(n_1 B_{12} - n_2 B_{21})}.$$

Divide numerator and denominator by $n_2 B_{21}$:

$$S_\nu = \frac{c}{4\pi} \frac{A_{21}/B_{21}}{(n_1/n_2)(B_{12}/B_{21}) - 1}.$$

Insert Boltzmann $n_1/n_2 = (g_1/g_2)e^{+h\nu_0/k_B T}$:

$$S_\nu = \frac{c}{4\pi} \frac{A_{21}/B_{21}}{(g_1 B_{12}/g_2 B_{21})e^{h\nu_0/k_B T} - 1}.$$

Apply $g_1 B_{12} = g_2 B_{21}$:

$$S_\nu = \frac{c}{4\pi} \frac{A_{21}/B_{21}}{e^{h\nu_0/k_B T} - 1}.$$

Insert $A_{21}/B_{21} = 8\pi h\nu_0^3/c^3$ (specific-intensity convention):

$$S_\nu = \frac{c}{4\pi} \cdot \frac{8\pi h\nu_0^3/c^3}{e^{h\nu_0/k_B T} - 1}.$$

Reduce:

$$\boxed{S_\nu = \frac{2h\nu_0^3}{c^2} \frac{1}{e^{h\nu_0/k_B T} - 1} = B_\nu(T).} \quad (13)$$

(c) Constant-coefficient solution

Write transfer equation as linear ODE:

$$\frac{dI_\nu}{dz} + \kappa_\nu I_\nu = j_\nu.$$

Multiply by integrating factor $e^{\kappa_\nu z}$:

$$\frac{d}{dz}(e^{\kappa_\nu z} I_\nu) = j_\nu e^{\kappa_\nu z}.$$

Integrate from 0 to z :

$$e^{\kappa_\nu z} I_\nu(z) - I_\nu(0) = \frac{j_\nu}{\kappa_\nu} (e^{\kappa_\nu z} - 1).$$

Divide by $e^{\kappa_\nu z}$, use $S_\nu = j_\nu/\kappa_\nu$:

$$\boxed{I_\nu(z) = I_\nu(0)e^{-\kappa_\nu z} + S_\nu(1 - e^{-\kappa_\nu z}).} \quad (14)$$

Beer's law as $\tau \rightarrow 0$; $I_\nu \rightarrow S_\nu$ as $\tau \rightarrow \infty$.

(d) Rosseland diffusion in slowly varying $T(z)$

Angle-resolved transfer equation, $\mu = \cos \theta$:

$$\mu \partial_z I_\nu(\mu) = -\kappa_\nu [I_\nu(\mu) - S_\nu].$$

Local LTE: $S_\nu = B_\nu(T(z))$. Diffusion expansion $I_\nu = I_\nu^{(0)} + I_\nu^{(1)} + \dots$ in $\kappa_\nu^{-1} \partial_z$.

Zeroth order, isotropic:

$$I_\nu^{(0)} = B_\nu(T).$$

First order, solve for $I_\nu^{(1)}$ from the LHS of the transfer equation:

$$I_\nu^{(1)}(\mu) = -\frac{\mu}{\kappa_\nu} \frac{\partial B_\nu}{\partial z} = -\frac{\mu}{\kappa_\nu} \frac{\partial B_\nu}{\partial T} \frac{\partial T}{\partial z}.$$

Define $F(z) \equiv \int I_\nu d\nu$ (specific intensity integrated over frequency, in the direction along z , $\mu = 1$):

$$F(z) = \int B_\nu(T) d\nu - \frac{\partial T}{\partial z} \int \frac{1}{\kappa_\nu} \frac{\partial B_\nu}{\partial T} d\nu.$$

Apply Stefan–Boltzmann $\int_0^\infty B_\nu d\nu = \sigma T^4 / \pi$ (per unit solid angle):

$$\int B_\nu d\nu = \frac{\sigma T^4}{\pi}.$$

Differentiate w.r.t. T :

$$\int \frac{\partial B_\nu}{\partial T} d\nu = \frac{4\sigma T^3}{\pi}.$$

Define Rosseland mean opacity:

$$\frac{1}{\kappa_R} \equiv \frac{\int_0^\infty (1/\kappa_\nu) \partial_T B_\nu d\nu}{\int_0^\infty \partial_T B_\nu d\nu}.$$

Rearrange to express the weighted integral:

$$\int \frac{1}{\kappa_\nu} \frac{\partial B_\nu}{\partial T} d\nu = \frac{1}{\kappa_R} \cdot \frac{4\sigma T^3}{\pi}.$$

Combine into F (and absorbing the convention that converts integrated intensity into the blackbody flux σT^4):

$$\boxed{F(z) = \sigma T^4 - \frac{4\sigma T^3}{\pi \kappa_R} \frac{\partial T}{\partial z}.} \quad (15)$$

Radiative analogue of Fourier’s law: “conductivity” $K_{\text{rad}} = 4\sigma T^3 / (\pi \kappa_R)$.

(e) Atom in equilibrium with a neutrino field

Two changes from the photon case:

- (i) Neutrinos are fermions: equilibrium occupation is Fermi–Dirac, $\bar{n} = 1/(e^{\hbar\omega_0/k_B T} + 1)$.
- (ii) Final-state Pauli blocking factor $(1 - \bar{n})$ for any neutrino emission.

Write absorption and emission rates per atom in mode of occupation $\bar{n}$:

$$R_{\text{abs}} = n_1 B_{12} \bar{n}, \quad R_{\text{em}} = n_2 [A_{21} + B_{21} \bar{n}] (1 - \bar{n}).$$

Note Pauli enforces $(1 - \bar{n})$ on *both* terms (spontaneous and stimulated).

Equate at equilibrium:

$$n_1 B_{12} \bar{n} = n_2 (A_{21} + B_{21} \bar{n}) (1 - \bar{n}).$$

Divide by $n_2(1 - \bar{n})$:

$$\frac{n_1}{n_2} B_{12} \frac{\bar{n}}{1 - \bar{n}} = A_{21} + B_{21} \bar{n}.$$

Apply Boltzmann $n_1/n_2 = (g_1/g_2)e^{\hbar\omega_0/k_B T}$ and use FD $\bar{n}/(1 - \bar{n}) = e^{-\hbar\omega_0/k_B T}$:

$$\frac{g_1}{g_2} B_{12} e^{\hbar\omega_0/k_B T} \cdot e^{-\hbar\omega_0/k_B T} = A_{21} + B_{21} \bar{n}.$$

Cancel exponentials:

$$\frac{g_1}{g_2} B_{12} = A_{21} + B_{21} \bar{n}.$$

LHS is T -independent, so RHS must be too. Match at all $\bar{n}$:

$$g_1 B_{12} = g_2 A_{21}, \quad B_{21} = 0.$$

$$g_1 B_{12} = g_2 A_{21},$$

$$B_{21} = 0 \text{ (no stimulated emission for fermions).}$$

(16)

Contrast with photon case: there A_{21}/B_{21} is set by the Planck cubic; here Pauli forbids stimulated emission so $B_{21} \rightarrow 0$, and A_{21} is directly fixed by absorption. Equilibrium spectrum is FD, not Planck.

Microscopic suppression: weak-vertex spontaneous neutrino emission $\sim G_F^2$ vs. EM $\sim \alpha^2$, a factor $\sim 10^{20}$; atomic spectra are EM, not weak.

Question 4 — Carbon levels, recoil and finite-nucleus shift in ^{208}Pb

Problem. (a) Term symbols of 0, 16.4, 43.4, 10192, 21648 cm^{-1} in ^{12}C ; (b) energy of a $1s^2 2s^2 2p 3s$ level decaying via $\lambda_1 = 193.09 \text{ nm}$, $\lambda_2 = 247.86 \text{ nm}$; (c) carbon foil under $\lambda_p = 462.6 \text{ nm}$ photons or $E_e = 2.68 \text{ eV}$ electrons; (d) recoil-shifted absorption ν ; (e) finite-nucleus shift in 1s of hydrogen-like ^{208}Pb , plus 2p and neutral-Pb 1s.

(a) Term symbols

Ground configuration $1s^2 2s^2 2p^2$. Pauli-allowed terms: 3P , 1D , 1S . Hund's rules: $^3P < ^1D < ^1S$; p^2 less than half full so smaller J lower.

$$\begin{array}{lll} 0 \text{ cm}^{-1} : ^3P_0, & 16.4 : ^3P_1, & 43.4 : ^3P_2, \\ 10192 : ^1D_2, & 21648 : ^1S_0. & \end{array}$$

(17)

(b) Energy of $1s^2 2s^2 2p 3s$ excited state

Photon energies via $E_\gamma = hc/\lambda$, $hc = 1240 \text{ eV nm}$:

$$E_{\gamma 1} = \frac{1240}{193.09}.$$

Evaluate:

$$E_{\gamma 1} = 6.42 \text{ eV}.$$

And:

$$E_{\gamma 2} = \frac{1240}{247.86} = 5.00 \text{ eV}.$$

Cascade $2p3s \rightarrow 2p^2(^1D) \rightarrow 2p^2(^3P)$, sum the two photon energies:

$$E_{\text{exc}} = 6.42 \text{ eV} + 5.00 \text{ eV} = 11.42 \text{ eV}.$$

Convert to wavenumbers via $8065.5 \text{ cm}^{-1}/\text{eV}$:

$$\tilde{\nu}_{\text{exc}} = 11.42 \text{ eV} \cdot 8065.5 \text{ cm}^{-1}/\text{eV}.$$

Evaluate:

$$\boxed{\tilde{\nu}_{\text{exc}} \simeq 9.21 \times 10^4 \text{ cm}^{-1}.} \quad (18)$$

(c) Photon vs. electron probe at 2.68 eV

Photon energy:

$$E_\gamma = hc/\lambda_p = \frac{1240}{462.6} = 2.681 \text{ eV}.$$

Wavenumber $\tilde{\nu} = 2.681 \cdot 8065.5 \simeq 2.16 \times 10^4 \text{ cm}^{-1}$, matching the $^3P \rightarrow ^1S_0$ excitation at $21\,648 \text{ cm}^{-1}$.

Photon probe. $^3P \rightarrow ^1S_0$ within $2p^2$: same parity ($\Delta\ell = 0$, parity even) and $\Delta S = 1$. E1 doubly forbidden; absorption suppressed except via M1/E2 (forbidden-line strength). Foil is essentially transparent.

Electron probe. Coulomb / exchange scattering: not bound by E1 selection rules; $\Delta S \neq 0$ and $\Delta\ell$ even allowed. Excites $^3P \rightarrow ^1S_0$ and $^3P \rightarrow ^1D_2$; subsequent slow forbidden fluorescence. Spectrum richer than photon case.

(d) Photon frequency with mass change ΔM

Conservation in lab frame, atom initially at rest; absorption gives the atom momentum $p = \hbar\omega/c$.

Energy + momentum invariants ($E_f^2 - p_f^2 c^2 = M_f^2 c^4$):

$$[\hbar\omega + Mc^2]^2 - (\hbar\omega)^2 = (M + \Delta M)^2 c^4.$$

Expand the square:

$$(\hbar\omega)^2 + 2\hbar\omega Mc^2 + M^2 c^4 - (\hbar\omega)^2 = (M + \Delta M)^2 c^4.$$

Cancel $(\hbar\omega)^2$:

$$2\hbar\omega Mc^2 + M^2 c^4 = (M + \Delta M)^2 c^4.$$

Solve for $\hbar\omega$:

$$\hbar\omega = \frac{(M + \Delta M)^2 - M^2}{2M} c^2.$$

Expand the numerator:

$$\hbar\omega = \frac{2M\Delta M + (\Delta M)^2}{2M} c^2.$$

Split into two terms:

$$\hbar\omega = \Delta M c^2 + \frac{(\Delta M)^2 c^2}{2M}.$$

Identify $\Delta E_0 = \Delta M c^2$:

$$\boxed{\hbar\omega = \Delta E_0 + \frac{(\Delta E_0)^2}{2Mc^2}.} \quad (19)$$

Second term is the recoil shift $p^2/2M$.

(e) Finite-nucleus shift in $1s$ of hydrogen-like ^{208}Pb

Uniform sphere of radius R , Gauss's law:

$$V_{\text{nuc}}(r) = \begin{cases} -\frac{Ze^2}{4\pi\epsilon_0} \cdot \frac{1}{2R} \left(3 - \frac{r^2}{R^2} \right), & r < R, \\ -\frac{Ze^2}{4\pi\epsilon_0 r}, & r \geq R. \end{cases}$$

Perturbation $\delta V = V_{\text{nuc}} - V_{\text{pt}}$ vanishes for $r \geq R$:

$$\delta V(r) = \frac{Ze^2}{4\pi\epsilon_0} \left[\frac{1}{r} - \frac{1}{2R} \left(3 - \frac{r^2}{R^2} \right) \right] \quad (r < R).$$

Approximation: $R \ll a_0/Z$ (numerically 7 fm vs. 6×10^{-13} m), so $|\psi_{1s}(r)|^2 \simeq |\psi_{1s}(0)|^2 = Z^3/(\pi a_0^3)$ inside.

First-order shift:

$$\Delta E = |\psi_{1s}(0)|^2 \int_0^R \delta V(r) 4\pi r^2 dr.$$

Insert δV and absorb 4π :

$$\int_0^R \delta V \cdot 4\pi r^2 dr = \frac{Ze^2}{\epsilon_0} \int_0^R \left[r - \frac{r^2}{2R} \left(3 - \frac{r^2}{R^2} \right) \right] dr.$$

Expand the integrand:

$$\int_0^R \delta V \cdot 4\pi r^2 dr = \frac{Ze^2}{\epsilon_0} \int_0^R \left[r - \frac{3r^2}{2R} + \frac{r^4}{2R^3} \right] dr.$$

Evaluate term by term:

$$\int_0^R r dr = \frac{R^2}{2}, \quad \int_0^R \frac{3r^2}{2R} dr = \frac{R^2}{2}, \quad \int_0^R \frac{r^4}{2R^3} dr = \frac{R^2}{10}.$$

Combine:

$$\int_0^R \delta V \cdot 4\pi r^2 dr = \frac{Ze^2}{\epsilon_0} \left[\frac{R^2}{2} - \frac{R^2}{2} + \frac{R^2}{10} \right] = \frac{Ze^2 R^2}{10\epsilon_0}.$$

Multiply by $|\psi_{1s}(0)|^2$:

$$\Delta E = \frac{Z^3}{\pi a_0^3} \cdot \frac{Ze^2 R^2}{10\epsilon_0} = \frac{Z^4 e^2 R^2}{10\pi\epsilon_0 a_0^3}.$$

Hydrogen-like ground state $|E_H| = Z^2 e^2 / (8\pi\epsilon_0 a_0)$, so $e^2 / (\pi\epsilon_0 a_0) = 8|E_H|/Z^2$:

$$\Delta E = \frac{Z^4 R^2}{10a_0^2} \cdot \frac{8|E_H|}{Z^2}.$$

Reduce:

$$\boxed{\Delta E = \frac{4}{5} Z^2 |E_H| \left(\frac{R}{a_0} \right)^2.} \quad (20)$$

Numerical estimate. Nuclear radius:

$$R = r_0 A^{1/3} = 1.2 \cdot 208^{1/3} \text{ fm} \simeq 7.1 \text{ fm}.$$

Ratio to Bohr radius:

$$\frac{R}{a_0} = \frac{7.1 \times 10^{-15} \text{ m}}{5.29 \times 10^{-11} \text{ m}} \simeq 1.34 \times 10^{-4}.$$

Square:

$$(R/a_0)^2 \simeq 1.80 \times 10^{-8}.$$

Hydrogen-like Pb ground state magnitude $|E_H| = Z^2 \cdot 13.6 \text{ eV}$:

$$|E_H| = 82^2 \cdot 13.6 \text{ eV} \simeq 9.14 \times 10^4 \text{ eV}.$$

Plug into the shift formula:

$$\Delta E = \frac{4}{5} \cdot 82^2 \cdot 9.14 \times 10^4 \text{ eV} \cdot 1.80 \times 10^{-8}.$$

Evaluate stepwise:

$$\frac{4}{5} \cdot 6724 = 5379.$$

$$5379 \cdot 9.14 \times 10^4 \text{ eV} = 4.92 \times 10^8 \text{ eV}.$$

$$4.92 \times 10^8 \text{ eV} \cdot 1.80 \times 10^{-8} \simeq 8.86 \text{ eV}.$$

Frequency shift via $\Delta\nu = \Delta E/h$:

$$\Delta\nu = \frac{8.86 \text{ eV}}{4.14 \times 10^{-15} \text{ eV s}}.$$

Reduce:

$$\boxed{\Delta\nu \simeq 2.1 \times 10^{15} \text{ Hz} \simeq 2 \text{ PHz.}} \quad (21)$$

(i) $2p$ in hydrogen-like Pb. $\psi_{2p} \propto r$ at small r , so $|\psi_{2p}(0)|^2 = 0$ and the leading term vanishes. Next-order shift goes as $\langle r^2 \rangle_{\text{in}}/a^5$; suppressed by extra $(R/a_0)^2 \sim 10^{-8}$. Essentially zero.

(ii) $1s$ in neutral Pb. The innermost $1s$ electron sees nearly the full bare Ze at small r (negligible inner-shell screening). $|\psi_{1s}(0)|^2$ reduced by a few percent (Pauli pressure of the second $1s$). The shift is the same $\sim 2 \text{ PHz}$ to within a few percent. Basis of $1s$ X-ray isotope-shift King plots in heavy elements.